

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

Abinaya Sri J(192524429)

Write a Prolog Program to Sum the Integers from 1 to n

AIM

To write a Prolog program to find the sum of integers from 1 to n.

OBJECTIVE

To calculate the sum of numbers from 1 to n using recursion in Prolog.

ALGORITHM

Step 1: Start.

Step 2: Define the base case as sum(0,0).

Step 3: Check whether N is greater than 0.

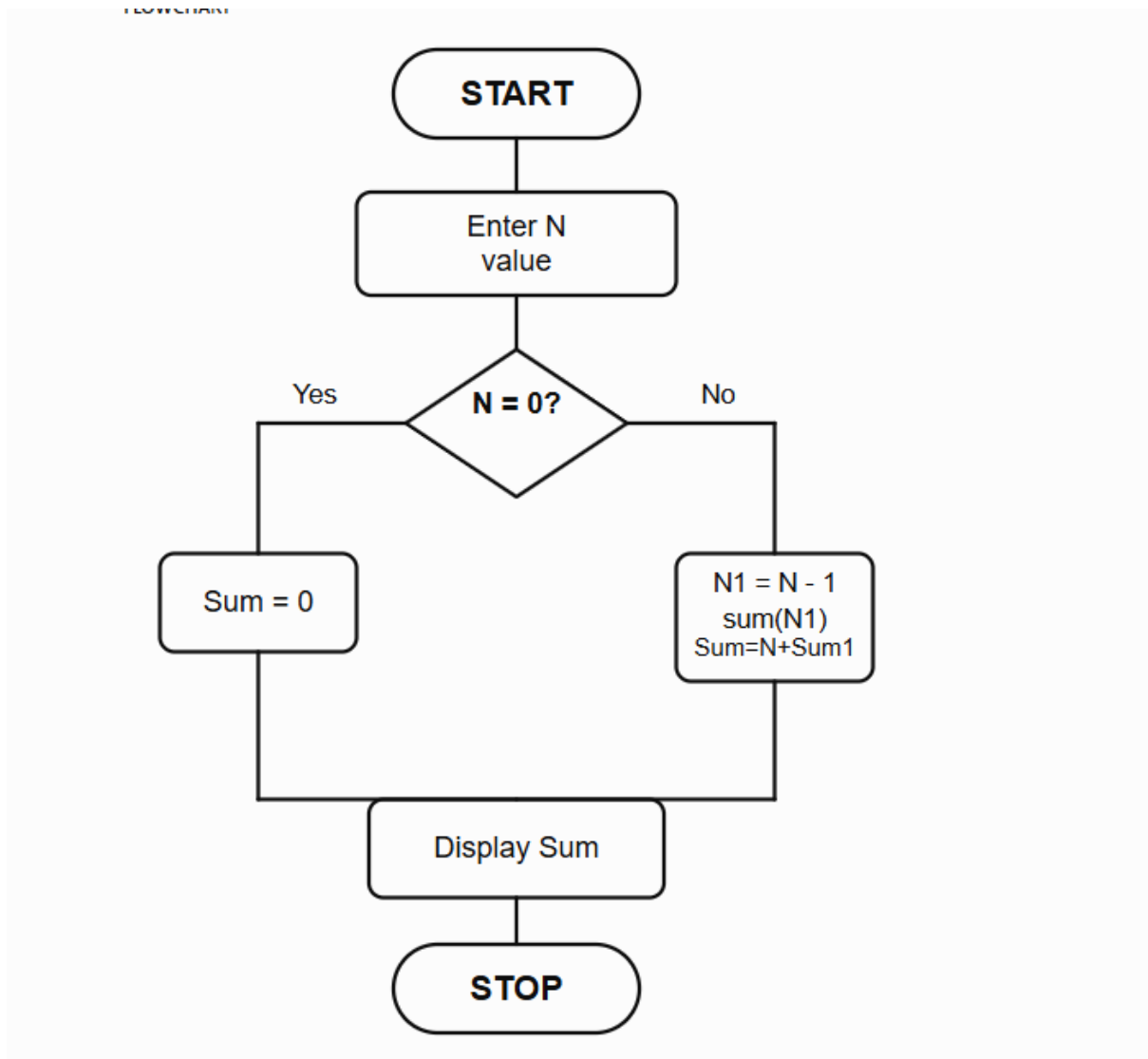
Step 4: Decrease N by 1.

Step 5: Recursively find the sum of N-1.

Step 6: Add N to the previous sum.

Step 7: Display the result. Step 8: Stop.

FLOWCHART



PROGRAM

sum(0,0). % base case: sum of 0 is 0

sum(N,Sum) :- N > 0, % recursive case: N is positive

N1 is N - 1, % decrement N

sum(N1,Sum1), % recursively compute sum of 1 to N-1

Sum is Sum1 + N. % add N to the sum of 1 to N-1

?- sum(5,Sum).

Sum = 15.

?- sum(10,Sum).

OUTPUT

```
?- sum(5,Sum).  
Sum = 15 .  
  
?- sum(10,Sum).  
Sum = 55 .
```

RESULT

Thus, the Prolog program to find the sum of integers from 1 to n was executed successfully and the output was verified.